

APSMO 2010 Set 5

1 Problems

Problem 1.1 In the diagram below, what is the sum of the numbers in the shaded boxes?

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24

Problem 1.2 A toll bridge charges \$4 for a car and \$6 for a truck. One day 200 of these vehicles crossed the bridge and paid a total of \$860 in tolls.

How many of these vehicles were trucks?

Problem 1.3 Zach has 2 blue lollies for every 1 red lolly. After he eats 1 of the blues and 2 of the reds, Zach has 5 blue lollies for every 2 red lollies.

How many lollies does Zach have at the start?

Problem 1.4 A cubical box without a top is 5cm on each edge. The box is filled with 125 identical 1cm cubes that exactly fill the box.

For how many 1cm cubes does exactly one face touch the box?

Problem 1.5 The average of 6 consecutive odd numbers is 50.

What is the least of these numbers?